

WEATHERWISE — AI Weather Intelligence Platform

Enterprise Micro-SaaS Software Architecture & Deployment Proposal

1. EXECUTIVE OVERVIEW

WeatherWise is a cloud-native Software-as-a-Service (SaaS) weather intelligence platform designed to convert complex meteorological telemetry into real-time, actionable consumer and commercial insights. Rather than delivering raw temperature values, WeatherWise leverages context-aware algorithmic processing to compute clothing requirements, outdoor activity viability metrics, and hyper-local environmental risk profiles.

2. PROJECT TEAM & ENGINEERING ROSTER

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COURSE UNIT

Web Application Engineering / SaaS Architecture

SUBMISSION DATE

September 2026

TARGET DEPLOYMENT

Serverless Cloud Edge (Vercel / AWS Lambda)

PROJECT LIFECYCLE

Phase 1 Production (MVP Complete)

SAAS CORE VALUE PROPOSITION

WeatherWise solves weather-data fatigue by providing instant automated recommendations. Designed with modular cloud microservices, it bridges consumer utilities and B2B API licensing models.

3. CORE FEATURES & FUNCTIONAL CAPABILITIES

Real-Time Weather Pipeline

Asynchronous integration with OpenWeatherMap REST API streaming temperature, wind speed, pressure, and cloud coverage.

Multi-City Context Engine

Preset global city switcher (Karachi, Lahore, Islamabad, Delhi, New York, London, Tokyo) plus instant dynamic lookups.

AI Contextual Advisory

Automated heuristics calculating real-time apparel choices and outdoor suitability based on thermal indices.

Predictive Extended Forecast

5-day environmental forecasting module engineered for user planning and weather risk mitigation.

4. SYSTEM TECHNOLOGY STACK

LAYER	TECHNOLOGIES	ARCHITECTURAL PURPOSE
Frontend UI	HTML5 CSS3 Flexbox	High-performance responsive interface, low render overhead, mobile-first design.
App Engine	Vanilla JS (ES6+)	Asynchronous fetch handling, state mutation, client caching, recommendation engine.
Data API	OpenWeather REST	Structured JSON weather telemetry streaming.
SaaS Deployment	Serverless CDN	Edge deployment ensuring sub-50ms latency globally.

5. MONETIZATION & SUBSCRIPTION STRATEGY

TIER	AUDIENCE	PRICING	INCLUDED FEATURES
Free Tier	Casual Users	\$0 / Month	Standard 5-day forecast, global city lookups, basic attire recommendations.
Pro SaaS	Power Travelers	\$9.99 / Month	Minute-by-minute rain radar, ad-free UI, automated severe weather alerts via SMS.
Enterprise	Logistics / Events	Custom Tier	Dedicated API endpoints, high-volume query access, custom analytical dashboards.

6. IMPLEMENTATION ROADMAP

- Phase 1 (Production MVP):** Weather data pipeline, rule-based AI recommendations, responsive dark-mode UI.
- Phase 2 (Near-Term):** User accounts (OAuth/Firebase), location bookmarking, push notifications.
- Phase 3 (Scale):** LLM integration for conversational weather queries and B2B enterprise dashboard.